

5a	Faraday's Law states that the induced e.m.f. in a circuit is <u>directly proportional</u> to the <u>rate of change of magnetic flux-linkage</u>	B2
	Lenz's Law states that the direction of the induced e.m.f. is such that the current it causes to flow opposes the change causing it.	B1
bi	$\omega = 2\pi f$ $= 314 \text{ rad s}^{-1}$	C1 A1
bii	Max emf = ωBA $= (314.15)(0.5)(0.09)$ $= 14.1 \text{ V}$	C1 A1
biii	Max induced current $= 14.13 / 12$ $= 1.2 \text{ A}$	C1 A1
biv	reduce loss of magnetic flux/maximize magnetic flux linkage between primary and secondary coils	A1
bv	Correct shape sinusoidal (starting from emf = 0) and 2 cycles Correct labels of max emf and period	B1 B1